

Multi-Relation Attention Network for Image Patch Matching

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Abstract—Deep convolutional neural networks attract increasing attention in image patch matching. However, most of them rely on a single similarity learning model, such as feature distance and the correlation of concatenated features. Their performances will degenerate due to the complex relation between matching patches caused by various imagery changes. To tackle this challenge, we propose a multi-relation attention learning network (MRAN) for image patch matching. Specifically, we propose to fuse multiple feature relations (MR) for matching, which can benefit from the complementary advantages between different feature relations and achieve significant improvements on matching tasks. Furthermore, we propose a relation attention learning module to learn the fused relation adaptively. With this module, meaningful feature relations are emphasized and the others are suppressed. Extensive experiments show that our MRAN achieves best matching performances, and has good generalization on multi-modal image patch matching, multi-modal remote sensing image patch matching and image retrieval tasks.

Index Terms—Image matching, feature relations, attention learning.

I. INTRODUCTION

IMAGE patch matching aims to measure the similarities between image patch-pairs, and determine whether these image patches were acquired from the same interest point or the same identification [1], [2]. It is very important in many applications of computer vision, such as multi-view stereo reconstruction [3], image registration [4], image retrieval [5], person re-identification [6], vehicle re-identification [7], object detection [8], video object segmentation [9] and text co-detection in multi-view scene images [10].

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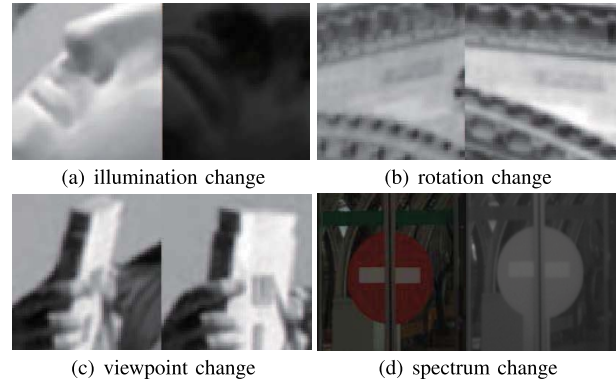


Fig. 1. There are great differences caused by illumination change, rotation change, viewpoint change and spectrum change in image patch-pairs.

The cores of image patch matching methods are comparing objects and ways of similarity measurement. As shown in Table I, the majority of traditional methods compute their correlation metrics based on statistical information of image pixels or gradient information. For example, some papers aim to search for the matched image patches with the maximal mutual information (MI) or normalized cross correlation (NCC). Other popular methods are based on handcrafted features, such as SIFT [11], SURF [12] and its variants [13], [14], which are related to the statistics of local gradient direction histograms. Meanwhile, their similarity depends on feature distance, such as Euclidean distance or Cosine distance. However, these traditional methods suffer in the case of significant differences between matching patches and lack robustness to image changes, such as illumination change, rotation transformation and viewpoint change.

Recently, with the development of deep learning, many matching methods proposed to use high-level features in place of handcrafted features. Under significant image changes, high-level features have stronger invariance and discrimination in comparison with existing handcrafted features. In consequence, deep learning methods have shown outstanding performances on image patch matching. There are two main streams to deep learning methods: Descriptor learning [15]–[21] and Metric learning methods [1], [22], [23]. The former focus on learning high-level features through convolutional neural networks and aim to reduce the distance between matching patches while pushing away the non-matching image patches in feature subspace. Metric learning methods perform both

TABLE I
COMPARISON AMONG DIFFERENT IMAGE PATCH MATCHING METHODS

Methods	Objects	Measurement mode	Merits	Demerits
Traditional methods				
Statistics of pixels	Image pixels	Mutual information (MI) Normalized cross correlation (NCC)	Simple computation	Less robust to image changes
Statistics of gradient (Low-level features)	Handcrafted features (SIFT, SURF, HOG)	Feature distance (Euclidean distance, Cosine distance)	Invariance to slight changes (Illumination invariance, Rotation invariance)	Less robust to significant image changes
Deep learning methods				
Descriptor learning	High-level features	Feature distance (Euclidean distance)	Strongly robust to image changes	Less effectiveness to complex image changes
Metric learning	Single feature relation (Feature concatenation, Feature difference)	Fully connected layers	Strongly robust to image changes	Less effectiveness to complex image changes
Our method	Fused feature relation Multiple feature relations	Fully connected layers	Strongly robust to image changes	Effectiveness to complex image changes

feature learning and similarity learning using one deep network. Specifically, the high-level features of patch-pairs are concatenated along the channel dimension (Feature concatenation, FC), subtracted by the element-wise mode (Feature difference, FD) and then input to several fully connected layers to predict their similarity score. Compared with the fixed feature distance measurement in descriptor learning methods, adaptive similarity learning mode in metric learning methods through an end-to-end method is more flexible, and can achieve better matching performances. However, as shown in Fig. 1, there are various changes between matching patch-pairs, such as viewpoint changes, illumination changes, geometrical transformation and image modal changes, which will lead to complex and diverse feature relations. Therefore, the single similarity learning model struggles to model this feature relation, resulting in performance degradation.

To solve this problem, we propose a multi-relation attention learning network (MRAN) for image patch matching, which can dramatically improve the generalization performances of matching networks. First, we propose to fuse multiple feature relations (MR) for similarity learning, including feature concatenation (FC), feature product (FP) and feature difference (FD). They are representative feature relations and measure the feature similarity from different perspectives. FC indicates adopting the channel-wise feature fusion mode for matching label prediction, FP captures the feature correlation for matching and FD computes the feature difference for matching. Thanks to the complementary advantages of different feature relations, MR has the capability to capture complex and diverse relations between the matching patch-pairs, which effectively enhances robustness to various changes. Additionally, in order to emphasize the meaningful feature relations and suppress the meaningless ones, we propose a relation attention module in MR learning network to learn the fused relation adaptively. In the training process, we compute matching loss for each feature relation and fused feature relation. Then, the acquired multiple losses are used for network optimization. Multiple losses will bring rich supervised

learning information from various perspectives, which can speed up the network convergence and boost matching performances. The contributions of this work are summarized as follows:

- First, we propose a novel multi-relation attention learning network (MRAN), which fuses multiple feature relations (MR) for image patch matching. With the complementary advantages of different feature relations, it can deal with complex and diverse image changes.
- Second, we propose a relation attention module (RA) to adaptively learn the fused relation, which forces the network to pay more attention to the meaningful feature relations and suppress the meaningless ones.
- Third, we propose to use multiple losses from different feature relations to train our network. Extensive experiments on the image patch matching benchmark datasets show that the proposed MRAN outperforms the state-of-the-art methods and demonstrate good generalization on multi-modal image patch matching and image retrieval task.

II. RELATED WORK

In this part, we will briefly introduce two classes of deep learning methods for image patch matching, descriptor learning and metric learning, and attention mechanism in various tasks.

A. Deep Matching Networks

1) *Descriptor Learning*: Descriptor learning networks are mainly used to learn high-level features. The similarity of patch-pairs depends entirely on their feature distance. Please refer to Fig. 2(a). The major modifications for descriptor learning come from loss function and sampling strategy. The widely-used loss functions are hinge embedding loss [24] and triplet loss [17]–[19]. Hinge embedding loss maximizes the feature distance of non-matching patch-pairs, while hopes the feature distance of matching samples is close to zero. Triplet

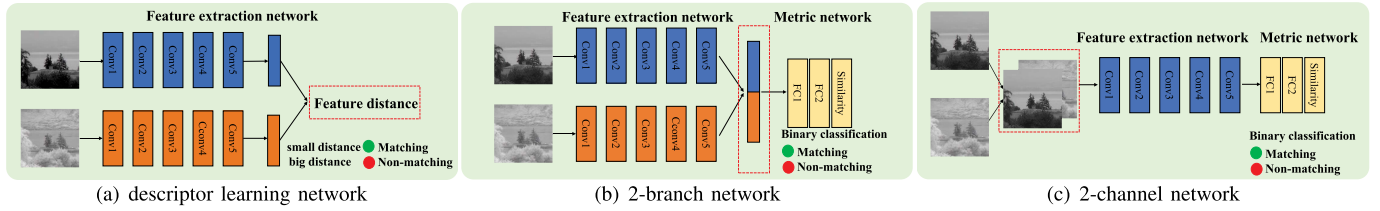


Fig. 2. Different deep network structures for image patch matching. (a) is a descriptor learning network, which aims to learn a feature vector for image patch and measures image similarity based on the feature distance. (b) is a metric learning network with 2-branch structure and (c) is a metric learning network with 2-channel structure, which use neural network to predict the matching labels directly.

loss pays more attention on the relative distance between matching samples and non-matching samples in a triplet. Moreover, Kumar *et al.* [18] introduced a global loss into the matching network, which aims to distinguish inter-class distribution and compact intra-class distribution. However, after a few iterations of training, the similarity of most samples can be predicted correctly. With very small loss value, these easy samples would not feedback effective information for training, which will lead to slower training speed and worse matching results. To solve this problem, many hard samples mining methods were proposed to exploit the informative training samples [15]–[21]. L2-Net [15] and HardNet [16] proposed to select the hardest negative sample among a mini-batch, and then feedback their loss information to optimize the network. Additionally, Zheng *et al.* [20] proposed to use generated hard samples rather than selected hard samples, which can greatly augment the training dataset and boost matching performances. However, hard samples mining methods overly concerned about the hard samples, but ignore the rich information available from other samples, which will lead to an unstable training process and even collapse.

2) *Metric Learning*: Both features and similarity are learned through deep neural networks in metric learning methods. Most of the improvements are related to network structure. As shown in Fig. 2 (b) and (c), the representative networks are 2-branch network [1] and 2-channel network [22]. Both of them conduct similarity learning by feature concatenation (FC) method, the former is based on raw pixels and the later is based on high-level features. Specifically, a 2-branch network has two feature extraction branches with the same structure and a metric subnetwork. One feature extraction branch for each patch of the input patch-pair. It connects the features of two patches along the channel dimension then inputs them to the metric subnetwork for feature similarity learning. A 2-branch network that shares weights between two feature extraction branches is called a Siamese network [1], [22], [23]. If the weights are not shared, it is a Pseudo-Siamese network [22], [23]. Therefore, the Pseudo-Siamese network has more flexibility than the Siamese network, which makes it more suitable for multi-modal images. However, the large number of parameters in Pseudo-Siamese network will increase the risk of over-fitting. A 2-channel network [22], [23] concatenates two raw image patches along channel dimension then predicts the matching label. After that, Quan *et al.* [2] proposed an AFD-Net to aggregate the feature difference (FD) from different feature levels, which computes feature relation by

element-wise mode and achieved faster and better performances. Subramaniam *et al.* [25] proposes a NCC-Net based on normalized cross correlation (NCC), feature difference and deep convolutional network for image patch matching. However, their NCC similarity computation needs to be performed multiple times in the neighbourhood region.

Different from descriptor learning methods relying on feature distance for similarity measurement, metric learning methods propose to learn similarity through deep networks, which are more flexible and effective at modelling feature relation and obtain better results. However, single similarity learning mode struggles to model the complex feature relations that are caused by various changes, leading to a decrease in matching performances. Moreover, there is little prior work focused on modifying the similarity learning model. To solve this problem, we propose to fuse multiple feature relations for similarity learning and further improve matching performances.

B. Attention Mechanism

The attention mechanism aims to emphasize the most informative components of an input signal and suppress the less useful parts [26]–[28]. Due to the powerful learning ability of the deep neural network and the effectiveness of attention mechanism, the combination of deep learning and attention mechanism has become a vigorous trend and has shown significant advantages in various tasks, including machine translation [28], image caption [29]–[31], semantic segmentation [32]–[35], image classification [36], image inpainting [37], human pose estimation [38], image super-resolution [39], action classification [40] and person re-identification [41]–[43].

Hu *et al.* [26] proposed a “squeeze-and-excitation” (SE) model to conduct feature recalibration by modelling the interdependencies between the channels of feature maps. Woo *et al.* [27] proposed a “convolutional block attention module” (CBAM) to perform attention learning along both channel dimension and spatial dimension of convolutional features, sequentially. It achieved significant improvement for image classification and detection. Zhang *et al.* [44] proposed to combine the self-attention model and generative adversarial network for image generation, which can model long range, multi-level dependencies across image regions. Huang *et al.* proposed the criss-cross attention network (CCNet) [34], [35] for capturing rich contextual information from full-image dependencies. Compared with the dense non-local attention module, CCNet had high computational

efficiency and lower computational resources. It achieved state-of-the-art performance on various semantic segmentation benchmarks. Lu *et al.* [30] proposed an adaptive attention model for image captioning with a visual sentinel, which determined whether the decoder uses image information or merely relies on the language model when generating the next word. Wang *et al.* [36] proposed a residual attention network to generate attention-aware features for image classification. Yu *et al.* [37] proposed to use the contextual attention module to select textures from background regions to generate the missing image patches. Zhang *et al.* [39] proposed a channel attention network for image super-resolution. Chen *et al.* [43] proposed a high-order attention module to utilize the high-order statistics of convolutional maps for person re-identification, which can increase the discrimination of attention proposals. However, few studies have been conducted on the attention mechanism for image matching. In this paper, we propose a relation attention module to learn the fused relation adaptively for image patch matching.

III. PROPOSED METHOD

In this paper, we propose a novel multi-relation attention learning network (MRAN) for image patch matching. The overall framework is shown in Fig. 3. The proposed network mainly contains three parts: features learning subnetwork, feature relations learning subnetwork, and fused feature relation attention learning subnetwork. Firstly, the feature learning subnetwork is used to extract features from image patches. Secondly, we compute the multiple feature relations, including feature concatenation (FC), feature difference (FD) and feature product (FP). Then, the fused multi-relation (MR) is adaptively learned by a relation attention module. Finally, more accurate matching labels can be inferred from the fused MR. More details are as follows.

A. Features Learning

Invariant and discriminative features are very important for matching, which determine the performances and the robustness of matching networks. In this paper, we use a 2-branch network that shares the network structure and weights to extract high-level features, as shown in the left of Fig. 3. For image patch-pair (p_1, p_2) , the extracted features can be formulated as,

$$\begin{aligned} f_1 &= F(p_1; \theta_F), \\ f_2 &= F(p_2; \theta_F), \end{aligned} \quad (1)$$

where F is the feature learning subnetwork, θ_F indicates its parameters, f_1 and f_2 are extracted high-level features of the input patches, respectively.

In the feature learning network, each branch contains four convolutional blocks, each of which has two convolutional layers with the kernel size of 3×3 , the activation function of Relu and the same number of kernels. Specifically, the numbers of kernels in four blocks are 32, 64, 128, 256, respectively, and the strides of the convolutional layers in each block are 1 and 2, respectively. Thus, the width and height of the feature map will be decreased by half through

every convolutional block. Following the same setting as [2], we also add a batch normalization (BN) layer after each convolutional layer to speed up the training process, and set an instance normalization (IN) layer after the first two convolutional blocks to remove the differences in low-level features.

B. Feature Relations Learning

According to the extracted high-level features, we compute various feature relations for subsequent fused relation learning. As shown in the middle of Fig. 3, we mainly consider three feature relations: feature concatenation (FC), feature difference (FD), and feature product (FP). They can be formulated as follows:

- **Feature concatenation:**

$$FC = R(f_1 \oplus f_2; \theta_{RC}), \quad (2)$$

where $\oplus$ denotes connecting features along the channel dimension, R represents feature relation learning subnetwork, θ_{RC} indicates its parameters for FC learning.

- **Feature difference:**

$$FD = R(|f_1 - f_2|; \theta_{RD}), \quad (3)$$

where $| - |$ represents the absolute difference (element-wise subtraction), θ_{RD} denotes the parameters of feature relation learning subnetwork for FD learning.

- **Feature product:**

$$FP = R(f_1 \otimes f_2; \theta_{RP}), \quad (4)$$

where $\otimes$ denotes Hardma product (element-wise multiplication), θ_{RP} indicates the parameters of feature relation learning subnetwork for FP learning.

As mentioned in Section II-A2, existing matching networks usually use FC mode for similarity learning, which connects high-level features along the channel dimension. It pays more attention to the feature relation between channels but ignores the relation on the spatial dimension. Therefore, FC can tolerate viewpoint changes, rotation transformations and other spatial deformations. However, due to lack of spatial relation learning, FC cannot capture more accurate similarity. In contrast, FD and FP compute the feature relation from the perspective of difference and correlation, respectively. They model the feature relation on both the channel and spatial dimension through element-wise mode, and provide more accurate similarity learning. On the surface, the element-wise mode will make FD and FP suffer from spatial deformations. However, high-level features come from the deep layer and are abstract and invariant. So, FD and FP can conduct similarity learning effectively on high-level features. Above all, FD and FP are more suitable for illumination change, image modal change and slight spatial transformation. Considering the negative impacts of great spatial deformations on element-wise

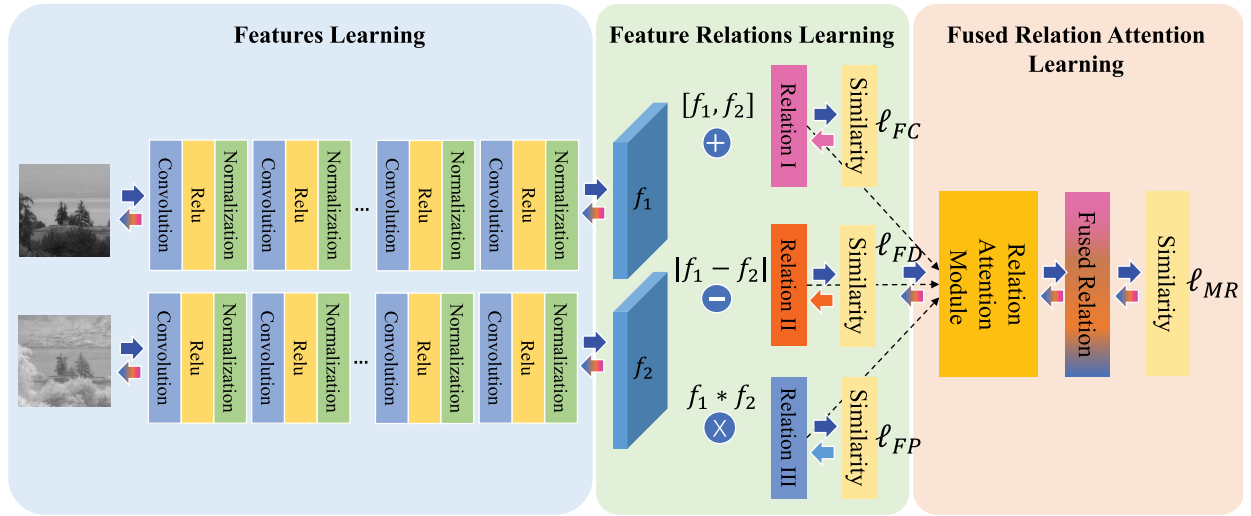


Fig. 3. Illustration of multi-relation attention learning network (MRAN), which has three parts: feature learning, feature relation learning, and fused feature relation attention learning. Moreover, multiple losses are used to optimize the network, which are computed from three different feature relations and the fused MR, respectively. The right arrow represents the feedforward process. The left arrow indicates backpropagation. There are multiple backpropagation information flows from multiples losses for network training.

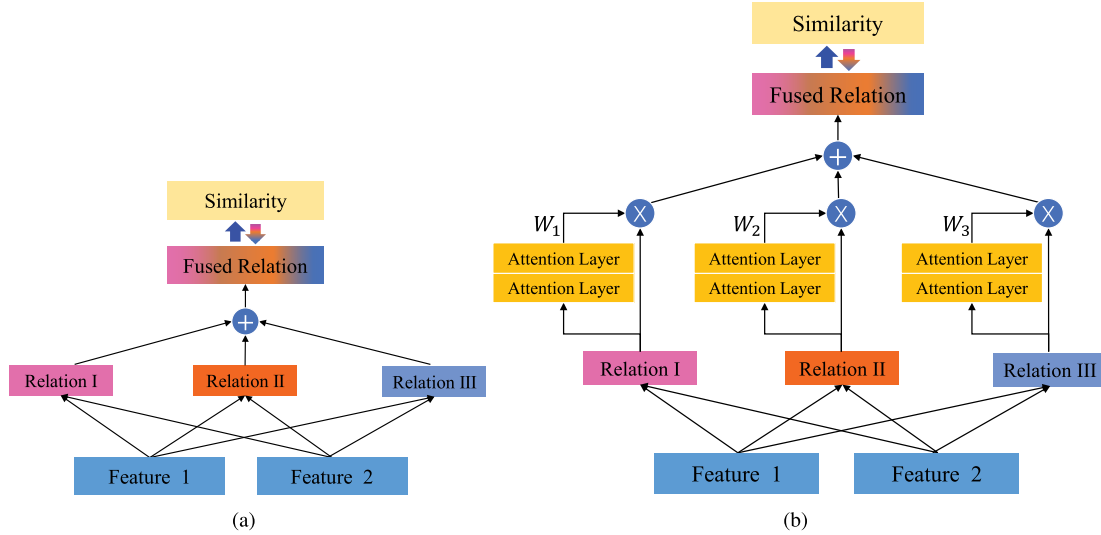


Fig. 4. Different MR learning networks. (a) without attention module (b) with a three-heads relation attention module.

learning mode, we propose to fuse FC, FD and FP to model complex and diverse feature relations caused by various kinds of changes.

C. Fused Feature Relation Attention Learning

In this part, we mainly fuse multiple feature relations (MR). The easiest method of fusion is concatenating different feature relations directly, as referred to Fig. 4(a). It can be formulated as:

$$MR = FC \oplus FD \oplus FP, \quad (5)$$

Based on this, different feature relations are treated equally without considering the characteristics of datasets. So, we propose a relation attention module to adaptively learn the fused MR. It aims to model the interdependencies among feature relations, and then emphasize the meaningful feature relations while suppressing meaningless ones.

As shown in Fig. 4(b), we propose a three-head relation attention module to conduct attention learning on three feature relations, separately, and acquire three heads of attentional relation, $Head_{FC}$, $Head_{FD}$ and $Head_{FP}$.

$$Head_{FC} = FC \otimes W_1, \quad (6)$$

$$Head_{FD} = FD \otimes W_2, \quad (7)$$

$$Head_{FP} = FP \otimes W_3, \quad (8)$$

where W_1 , W_2 and W_3 indicate the attention weights for three feature relations, respectively. They are computed by two fully connected layers with Relu and sigmoid activation functions, respectively.

$$W_1 = \text{sigmoid}(\text{Relu}(FC)), \quad (9)$$

$$W_2 = \text{sigmoid}(\text{Relu}(FD)), \quad (10)$$

$$W_3 = \text{sigmoid}(\text{Relu}(FP)), \quad (11)$$

Finally, the fused MR learned by three-head relation attention module is:

$$MR = Head_{FC} \oplus Head_{FD} \oplus Head_{FP}, \quad (12)$$

D. Optimization

As previously mentioned, we can acquire three single feature relations, FC , FD , FP , and the fused multiple feature relation MR . All of them can be used to compute the similarity of patch-pairs:

$$\begin{aligned} S_\Phi &= M_\Phi(\Phi; \theta_{M_\Phi}), \\ \Phi &\in \{FC, FD, FP, MR\}, \end{aligned} \quad (13)$$

where S_Φ is the predicted similarity score based on the relation of Φ , M_Φ denotes the corresponding similarity learning layers, θ_{M_Φ} represents its parameters.

Then, we can infer the matching label of patch-pairs:

$$\hat{y}_\Phi = \text{softmax}(S_\Phi), \quad (14)$$

where $\hat{y}_\Phi$ is the predicted label based on relation Φ .

Finally, the matching loss ℓ_Φ can be computed based on the large margin cosine function [45].

$$\begin{aligned} \ell_\Phi &= -\frac{1}{N} \sum_{i=1}^N \log \frac{e^{s(\cos(\theta_{y_i}, i) - m)}}{e^{s(\cos(\theta_{y_i}, i) - m)} + \sum_{j \neq y_i} e^{s(\cos(\theta_j, i))}}, \\ W &= \frac{W^*}{\|W^*\|}, \quad x = \frac{x^*}{\|x^*\|}, \quad \cos(\theta_j, i) = W_j^T x_i, \end{aligned} \quad (15)$$

where N denotes the number of training samples, y_i is the ground-truth matching label of the i_{th} sample, x_i is the corresponding feature vector of the last fully connected layer, W_j represents the weight vector of the i_{th} label, θ_j is the angle between the W_j and x_i , s is the scale parameter, m is the cosine margin.

Specifically, there are four different loss functions ℓ_{FC} , ℓ_{FD} , ℓ_{FP} and ℓ_{MR} . All of them can be used for training, and the whole loss is formulated as:

$$\begin{aligned} L &= \sum_{\Phi} \ell_\Phi, \\ &= \ell_{FC} + \ell_{FD} + \ell_{FP} + \ell_{MR}. \end{aligned} \quad (16)$$

As shown in Fig. 3, multiple loss (ML) functions means various backward information flows for feature subnetwork training, which will dramatically speed up the training process and improve matching performances compared with the single loss (SL) of ℓ_{MR} . This point is supported by the experimental results in section IV-C.

IV. EXPERIMENT

A. Datasets

In experiments, we test our proposed MRAN on a single modal image patch matching dataset, Brown, and test its generalization on two multi-modal image patch matching datasets, VIS-NIR and VIS-LWIR. To further evaluate the performance of our proposal, we present its matching results on the multi-modal remote sensing image patch matching dataset optical-SAR. We also show the performances of MRAN on the image retrieval dataset, Romepatches.

1) *Brown*: Brown dataset has three subsets: Liberty, Notre Dame, and Yosemite, which are widely used for matching and local descriptor evaluation [1], [2], [15], [22], [46], [47]. Each subset contains 100K, 200K, and 500K patch-pairs, respectively. The patch size is 64×64 . As shown in Fig. 5, there are various changes between matching patches, such as illumination changes, rotation changes, scale changes and perspective changes. So, the challenge is that there are significant differences between matching patches. Similar to previous studies, we will train our framework on one subset and test on the other two.

2) *VIS-NIR*: VIS-NIR patch dataset is a benchmark dataset for multi-modal image patch matching evaluation [2], [23], [47]–[49], one mode is related to the visible spectrum (VIS) and the other is near-infrared (NIR), as shown in Fig. 6. VIS-NIR has nine subsets: Country, Field, Forest, Indoor, Mountain, Oldbuilding, Street, Urban and Water, and the number of each subset is listed in Table II. The size of the image patch is 64×64 , which was cropped from VIS and NIR images centred on SIFT points. Following the previous studies [23], [47], [48], our framework is also trained on the Country subset and tested on other subsets. It is worth noting that there are significant differences between the categories, including various natural scenery (tree, river, field and mountain) and multifarious human-made objects (building, car, road signs, furniture and advertising board). Therefore, it is very difficult to obtain a good generalization performance on all test categories.

3) *VIS-LWIR*: Since the spectral band between visible images and long-wave infrared (LWIR) images is farther than that of VIS-NIR, their appearance differences are more obvious. VIS-LWIR has 44 pairs of registered VIS images and corresponding LWIR images [50]. Following the same setting as VIS-NIR, we also extract patch-pairs with the size of 64×64 centred on SIFT points. The patches with the same SIFT point are matching, otherwise, they are non-matching. We take half of the patch-pairs for training and the other half for testing.

4) *Optical-SAR*: In the remote sensing image process, the combination of optical images and synthetic aperture radar (SAR) images is very important. Multi-modal optical and SAR image matching is the crucial task for multi-modal remote sensing image applications. We generate optical-SAR image patch-pairs from data SEN1-2 [51], [52]. It contains a lot of optical images and corresponding SAR images at different seasons, such as spring, summer, autumn and winter. The optical image and SAR image have a size of 256×256 and a resolution of 10m. As shown in Fig. 7, we can see that there are significant difference between optical images and SAR images caused by different imaging mechanisms. We also crop the local image patch around SIFT points from optical images and SAR images. The size of image patches is 64×64 . 583180 image patch-pairs are used for training and another 248274 image patch-pairs are used for network testing.

5) *Romepatches*: Romepatches is the benchmark dataset for image retrieval. It has 20,000 patches belong to 2,000 3D points. As shown in Fig. 8, each point has 10 patches. One patch is taken as the query, and the other patches with the

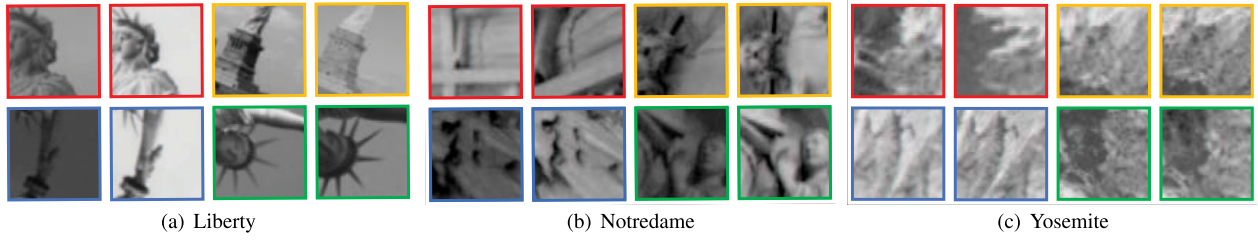


Fig. 5. Image patch-pairs of three subsets in Brown. The patches with the same color border are matching.

TABLE II
THE NUMBER OF IMAGE PATCH-PAIRS IN VIS-NIR DATASET

Category	Country	Field	Forest	Indoor	Mountain	Oldbuilding	Street	Urban	Water
Number	277504	240896	376832	60672	151296	101376	164608	147712	143104

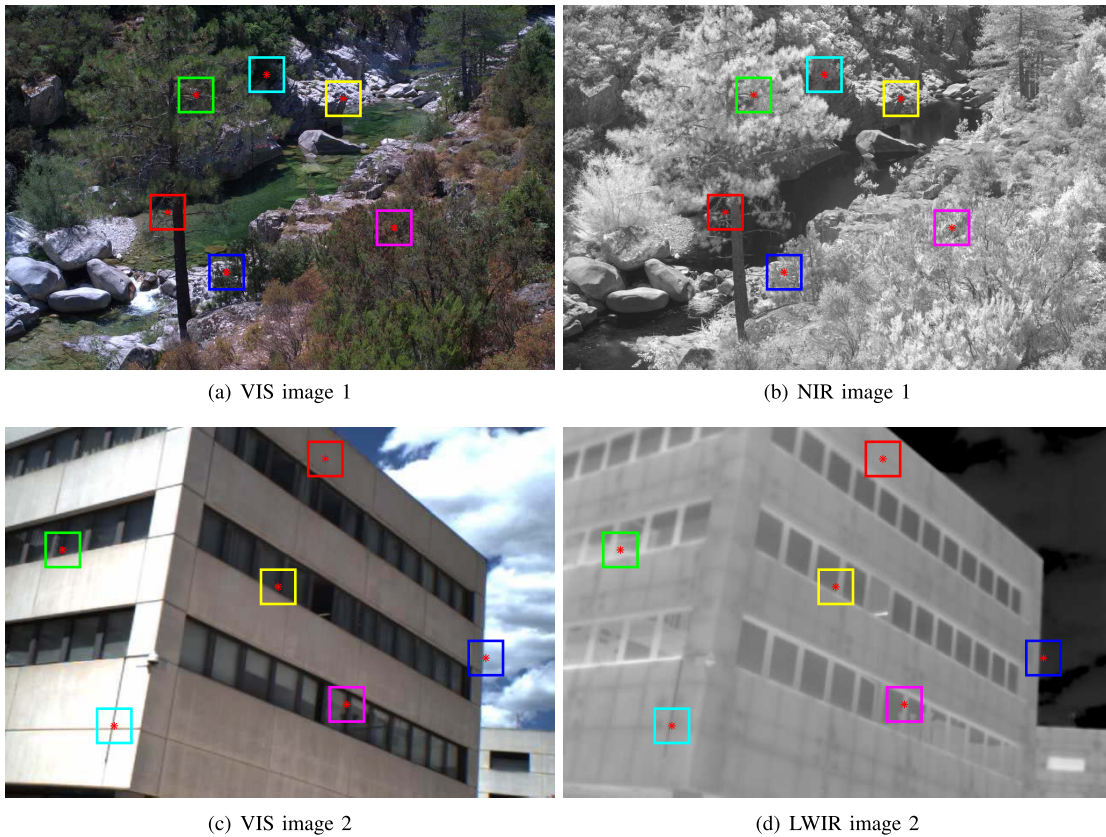


Fig. 6. (a) and (b) are Visible image (VIS) and corresponding near-infrared image (NIR). The patches with the same color border are matching. (c) and (d) are VIS and corresponding long-wave infrared image(LWIR). The wavelength range of them as follows: VIS: 0.4-0.7 μm , NIR: 0.75-1.4 μm , and LWIR: 8-15 μm . The spectral band between VIS-LWIR is much farther than that of VIS-NIR. So, their appearance differences are more obvious.

same point ID are targets. Half of the points are used to form training dataset, and the other half for building test dataset. The patches are resized from 51×51 to 64×64 by bilinear interpolation. In the training process, we build 300K image patch-pairs (100K matching and 200K non-matching). Mean average precision (mAP) will be reported on Romepatches, and a higher mAP means better retrieval performances.

B. Training and Testing Details

In training processing, we use stochastic gradient descent (SGD) to optimize our network, with a mini-batch

of 256 and an initial learning rate of 0.01, decaying by the factor of 0.9 every 5 epoches. Additionally, we conduct data augmentation by random flipping, rotating and contrast changes. The parameters in loss function are $s = 20$ and $m = 0.25$.

Following the evaluation protocol of [2], [15], [16], [23], [47], [48], we will present the false positive rate at 95% recall (FPR95) on each test subset, as well as the mean results and variance across all scenarios. The smaller FPR95 indicates better matching performances. Meanwhile, the mean result represents the average performance on all scenarios and shows the generalization of the matching network. Variance indicates

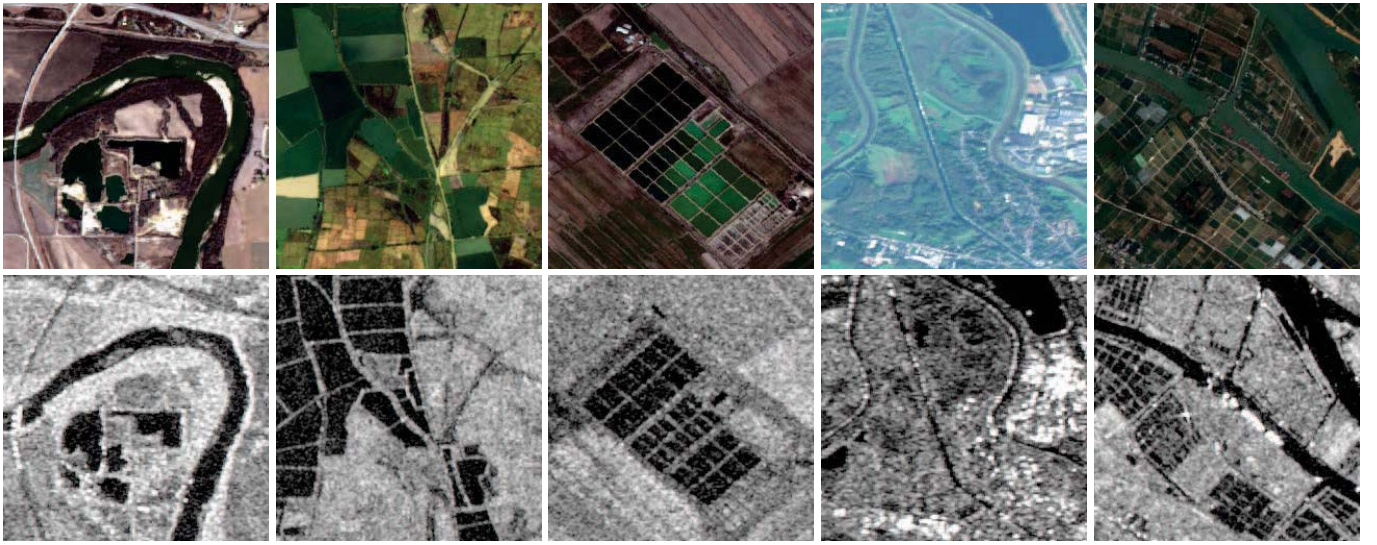


Fig. 7. Multi-modal remote sensing image pairs. The first line presents optical images and the second line shows SAR images.

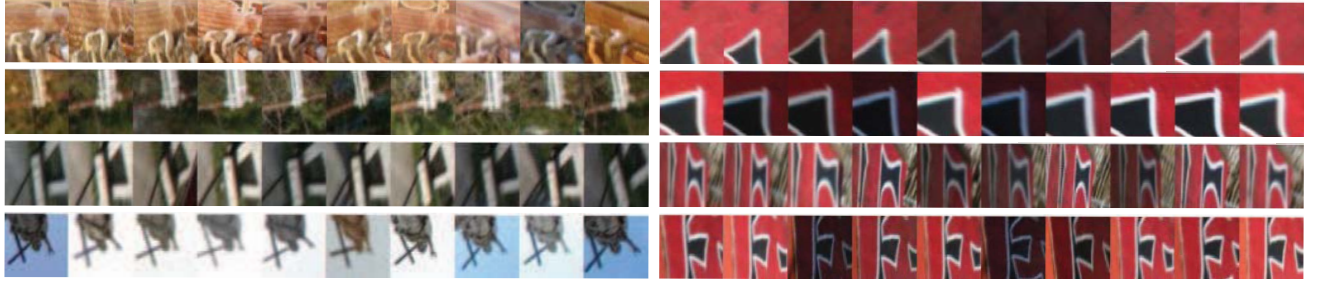


Fig. 8. Image patches in Rompatches dataset. Each point ID has 10 patches. The first patch is the query, other nine patches are retrieval targets.

TABLE III

FPR95 RESULTS OF MULTI-RELATION (MR) NETWORK AND SINGLE RELATION LEARNING METHODS (FC, FD, FP), SL DENOTES SINGLE LOSS OF ℓ_{MR} , ML REPRESENTS USING MULTIPLE LOSSES FOR TRAINING, “MR_3A” MEANS APPLYING THREE-HEAD RELATION ATTENTION MODULE

Training	Notredame	Yosemite	Liberty	Yosemite	Liberty	Notredame	Mean	Var
Test	Liberty		Notredame		Yosemite			
FC	4.96	5.32	1.63	1.67	3.25	3.96	3.47	2.51
FP	3.19	4.72	0.91	1.79	3.26	2.68	2.76	1.73
FD	2.25	3.67	0.78	1.03	2.30	2.59	2.10	1.13
MR_SL	1.83	2.94	0.59	0.90	2.08	2.31	1.78	0.78
MR_ML	1.70	2.65	0.50	0.85	1.75	1.97	1.57	0.61
MR_3A	1.47	2.09	0.50	0.77	1.69	1.75	1.38	0.38

the robustness of the network, and a smaller variance means a more robust matching network. In addition, we also report matching accuracy, and higher accuracy guarantees better performance.

C. Ablation Study

In this section, we evaluate the effectiveness of the proposed multiple feature relations learning (MR) and the relation attention module (RA).

1) *Influence of MR*: To demonstrate the influence of MR for patch matching, we compare it with different single feature relation (SR) learning methods on Brown dataset, including feature concatenation (FC), feature product (FP) and feature

difference (FD). Comparison results in terms of FPR95 are shown in Table III.

One can see that the proposed MR can significantly improve the matching performances over other SR methods on every test subset, even without multiple losses for optimization and attention module (MR_SL). Meanwhile, MR achieves a smaller variance, which shows the robustness and stability of MR. Moreover, multiple losses (ML) have rich backward information for training. It can speed up the training process and further enhance matching performances. Specifically, compared with FC, FP, and FD, the mean of FPR95 was increased 54.76%, 43.12%, and 25.24% by MR_ML, respectively. So, MR_ML encourages feature networks to extract more discriminative features and further helps MR to acquire a lower

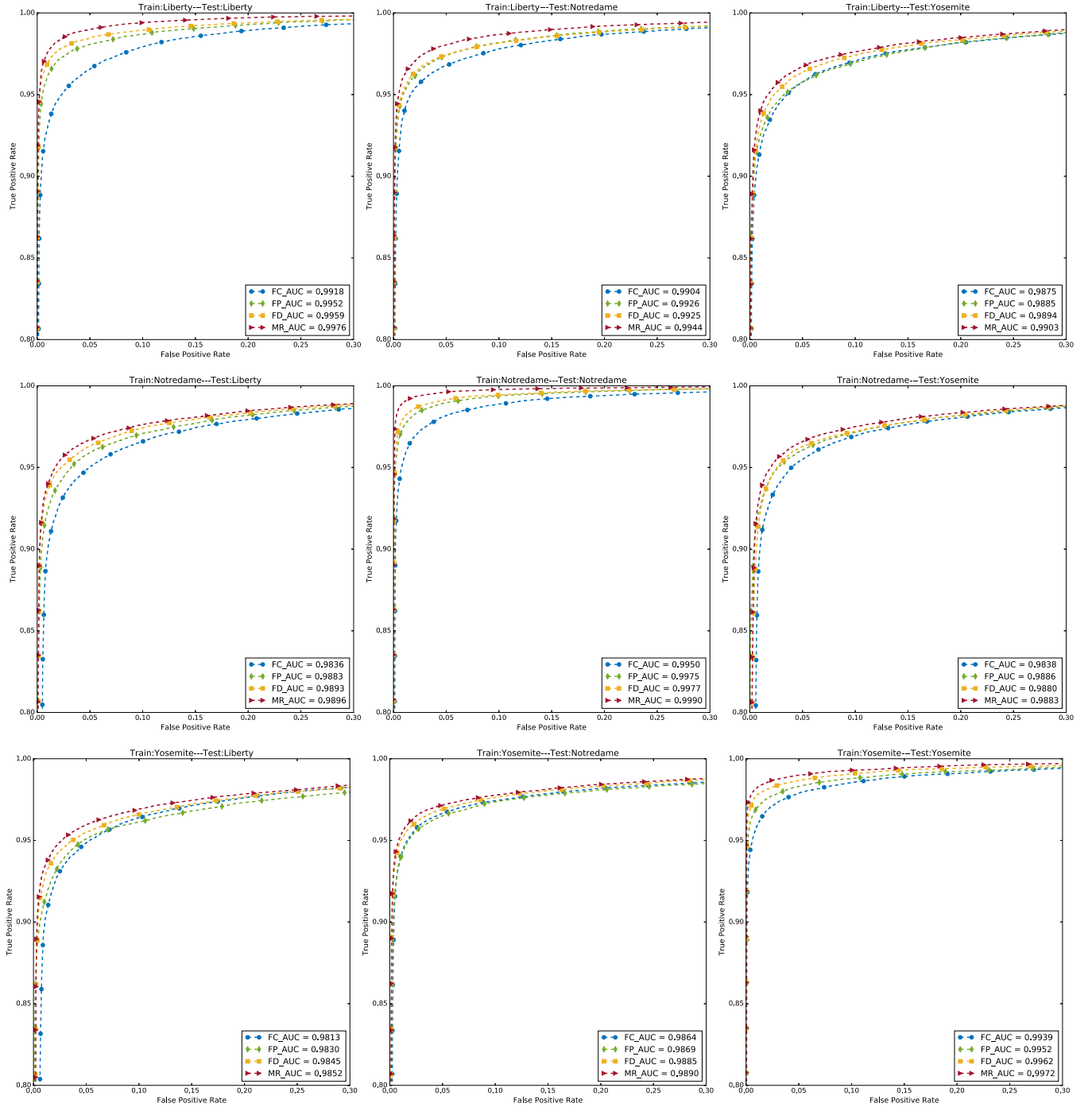


Fig. 9. The receiver operating characteristic curves of multi-relation learning network and various single feature relation networks on Brown dataset.

FPR95. Figure 9 shows the receiver operating characteristic (ROC) curves of MR and other single relation learning networks, which also demonstrates the advantages of multiple feature relations learning. Figure 10 displays the FPR95 changes during training process. It can be seen that MR can converge quickly and perform better than single relation learning methods on each test scenario. These results also prove the effectiveness of MR on image patch matching tasks.

2) *Influence of RA*: Since we propose a relation attention module in MR learning network (three-head relation attention learning “MR_3A”), we would like to evaluate

its contributions in our framework. The results are listed in Table III.

One can see that the relation attention module can further improve the matching performance on all test subsets and decrease the mean and variance of FPR95. These observations demonstrate the effectiveness of the attention mechanism. Specifically, MR_3A increases the average matching performances by 12.10%. Moreover, Fig. 10 also shows the relation attention module has the advantage in terms of matching performances, convergence speed, and training stability on most test scenarios.

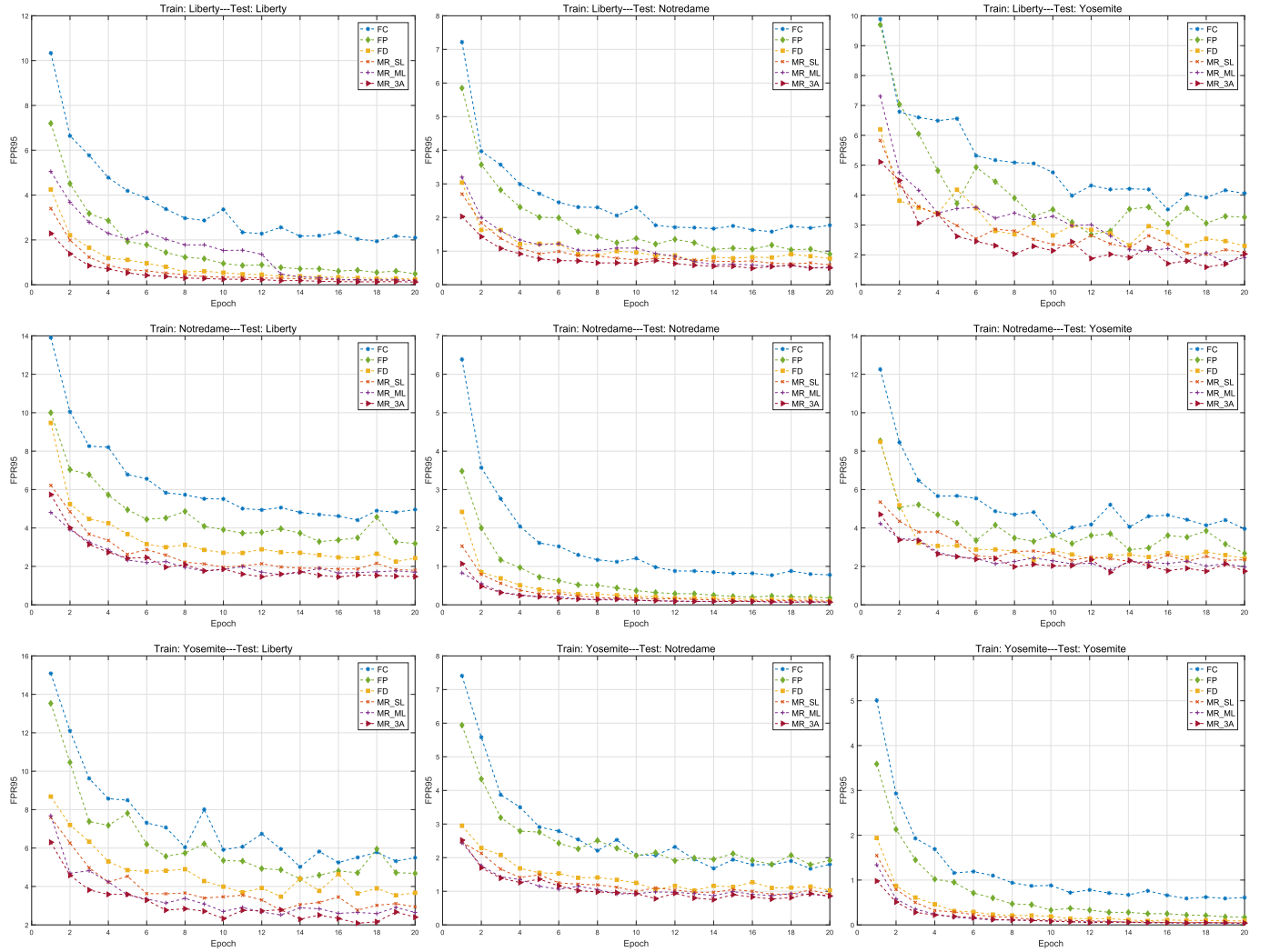


Fig. 10. The training efficiency of multi-relational attention learning network and various single feature relation learning networks on Brown dataset.

D. Comparison With Other Methods

To demonstrate the effectiveness of our proposal on image patch matching, we compare it with eighteen state-of-the-arts on Brown dataset. The results are listed in Table IV. From these experimental results, we can draw the following conclusions.

Firstly, due to the data-driven and multi-layered non-linear learning, deep learning methods raise the matching results of traditional methods by a large margin. Secondly, the modified deep matching methods from the perspectives of network structures, loss functions and hard samples mining can further improve matching performances. For example, 2-channel structure matching network of 2ch-deep [22] has better results than two branches network of MatchNet [1]. Additionally, the two-stream learning structure simultaneously focus on comparing the center parts and the surrounding parts of image patches, which also has a positive impact on matching. On the other hand, a suitable and effective loss function will promote the training process of the matching network and result in better results. For example, due to the additional constraint of global loss, TNet-TGLoss [18] improves the mean

results of TNet-TLoss [18] from 6.84% to 6.47%. Moreover, L2-Net [15] and HardNet [16] used the hard samples mining strategy to select more informative training samples for network optimization and achieved significant performances.

Additionally, different from the aforementioned directions of improvement in existing matching networks, our proposed method aims to promote matching performance from the view of various feature relations learning. It can be seen that MRAN has the smallest mean and variance of FPR95, which demonstrate our proposed MRAN outperforms state-of-the-art methods on image patch matching and has strong robustness. Specifically, compared with the handcrafted features RootSIFT, the mean value is raised 94.72% by MRAN. Moreover, MRAN significantly improves the mean matching results of MatchNet [1], 2ch-deep [22] and 2ch-2stream [22] about 81.45%, 67.61%, and 67.06%, respectively. It is worth noting that our proposal also performs better than the triplet loss network of TFeat-margin [19] and the mean value of FPR95 is decreased by 78.67%. Additionally, compared with L2-Net and HardNet, their mean performances are improved 37.84% and 8.61% by MRAN, respectively.

TABLE IV

THE COMPARISON FPR95 RESULTS AMONG OUR PROPOSAL AND EIGHTEEN STATE-OF-THE-ART METHODS ON BROWN DATASET. THE BOLD FONT INDICATES THE BEST PERFORMANCES. THE ITALICS DENOTES THE SUBOPTIMAL RESULTS

Training Test	Notredame Liberty	Yosemite	Liberty Notredame	Yosemite	Liberty Yosemite	Notredame	Mean	Var
RootSIFT [53]	29.65		22.06		26.71		26.14	14.65
L-BGM [54]	18.05	21.03	14.15	13.73	19.63	15.86	17.08	8.87
Convex optimization [55]	12.42	14.58	7.22	6.82	11.18	10.08	10.38	9.04
DeepDesc [24]	10.90		4.40		5.69		6.99	11.84
MatchNet [1]	6.90	10.77	3.87	5.67	10.88	8.39	7.44	7.89
2ch-deep DA [22]	4.55	7.40	2.01	2.52	4.75	4.38	4.26	3.66
2ch-2stream DA [22]	4.85	7.20	1.90	2.11	5.00	4.10	4.19	3.95
PN-Net [17]	8.13	9.65	3.71	4.23	8.99	7.21	6.98	6.16
Q-Net DA [48]	7.64	10.22	4.07	3.76	9.34	7.69	7.12	7.15
TFeat-ratoin [19]	8.07	9.53	3.47	4.23	8.53	7.24	6.84	5.99
TFeat-margin [19]	7.22	9.79	3.12	3.85	7.82	7.08	6.47	6.37
TNet-TLoss DA [18]	10.77	13.90	4.47	5.58	11.82	10.96	9.58	13.82
TNet-TGLoss DA [18]	9.91	13.45	3.91	5.43	10.65	9.47	8.80	12.40
SNet-Gloss DA [18]	6.39	8.43	1.84	2.83	6.61	5.57	5.27	6.17
L2-Net DA [15]	2.36	4.70	0.72	1.29	2.57	1.71	2.22	1.93
HardNet DA [16]	1.49	2.51	0.53	0.78	1.96	1.84	1.51	0.56
SCFDM DA [47]	1.47	4.54	1.29	1.96	2.91	5.20	2.89	2.70
AFD-Net DA [2]	1.53	2.31	0.47	0.72	1.63	1.88	1.42	<i>0.49</i>
MR_3A	1.47	2.09	<i>0.50</i>	<i>0.77</i>	<i>1.69</i>	<i>1.75</i>	1.38	0.38

Although our proposal does not adopt the hard samples mining strategy like L2-Net [15] and multi-level feature information fusion in AFD-Net [2], it also obtains the best matching results. MRAN can be adapted to different image changes, because it aggregates various feature relations for matching. So, our proposed MRAN is superior to existing methods on image patch matching.

E. Generalization on Multi-Modal Images

To demonstrate the generalization of our proposal, we test MRAN on multi-modal image patch matching, VIS-NIR and VIS-LWIR. In contrast to Brown dataset, there are significant appearance differences between multi-modal images (see Fig. 6). Taking VIS-NIR as an example, the matching results of different relation learning networks are presented in Table V. From these results we can see that FD has the best results of the single-FR learning methods, and FP comes second. Multiple feature relations improve image patch matching performance compared with single feature relation learning. FD + FP achieves the best performance of the two-relation learning methods. Specifically, FD + FC decrease the FPR95 results of FC from 1.88 to 1.34. Due to the joining of FP, the mean FPR95 result of FC is decreased from 1.88 to 1.56. Moreover, the combination of FD and FP boost the matching performance of FP from 1.65 to 1.32 and enhance the matching results of FD from 1.48 to 1.32. MR (FC + FP + FD) achieves better performances than two-relation learning networks. It has the smallest mean and variance in terms of FPR95. These results demonstrate the effectiveness of MR and relation attention module on multi-modal images.

We also compare the proposed MRAN with other representative deep matching methods on multi-modal images, such as 2-channel network [23], Siamese network [23], Pseudo-Siamese network, [23] and the central-surround two-stream networks (2channel-2stream, Siamese-2stream) [22]. Their matching accuracy changes versus training epoch are shown

in Fig. 12 and Fig. 11. It can be seen that two-stream networks perform better than single-stream networks. However, the significant differences between multi-modal images severely impair their matching performances, especially on VIS-LWIR. Fortunately, our proposed MRAN has the fastest convergence speed, the most stable training process, and the best matching performances on both VIS-NIR and VIS-LWIR. Therefore, MRAN has a good generalization performance on multi-modal image patch matching.

1) *Discussion*: Experimental results show that MRAN achieves the best performances on both single-modal and multi-modal image patch matching. However, the advantages of relation attention module on VIS-NIR are not as significant as on Brown. This is because VIS-NIR mainly has spectral differences, but no rotation changes or viewpoints changes. Thus, the changing pattern between matching patches is single and their feature relation is simpler, which means MRAN cannot take full advantages of multiple relation learning and attention mechanism. Therefore, MRAN is more suitable for the datasets with multiple changes.

2) *The Influence of Image Patch Size*: To present the influence of image patch size, we test our proposal with various image patch sizes on VIS-NIR dataset, such as 16×16 , 32×32 and 64×64 . As shown in Table. VI, as the image patch size increases, the matching accuracy at each test scenario gets better. Large image patches contain more image content, which will bring rich discriminating information for matching and enhance matching performance. However, they have more expensive computation and memory cost. Conversely, small image patches can speed up the computation process but lack representative features for matching, which will result in many false matching patch-pairs and degenerate matching performance. In most public local image patch datasets, the size of image patches is 64×64 . So, we also cropped the local image patch with the size of 64×64 in our experiments.

TABLE V

FP95 RESULTS OF DIFFERENT DEEP MATCHING NETWORKS ON VIS-NIR DATASET, INCLUDING SINGLE RELATION LEARNING METHODS (FC, FD, FP), TWO-RELATION FUSION LEARNING NETWORK (FC + FD, FD + FP, FC + FP) AND MULTI-RELATION (MR, FC + FD + FP) NETWORK. SL DENOTES SINGLE LOSS OF ℓ_{MR} , ML REPRESENTS USING MULTIPLE LOSSES FOR TRAINING, “3A” MEANS APPLYING THREE-HEAD RELATION ATTENTION MODULE IN MR NETWORK, “2A” REPRESENTS USING TWO-HEAD RELATION ATTENTION MODULE IN TWO-RELATION FUSION LEARNING NETWORK

Setting	Field	Forest	Indoor	Mountain	Oldbuilding	Street	Urban	Water	Mean	Var
FC	6.41	0.21	1.44	1.60	0.98	1.14	0.50	2.77	1.88	3.95
FP	5.64	0.11	1.29	1.48	0.79	0.80	0.63	2.47	1.65	3.09
FD	5.38	0.11	1.32	0.88	0.85	0.59	0.42	2.26	1.48	2.91
MR_SL	4.94	0.10	1.26	1.26	0.78	0.91	0.31	2.29	1.48	2.40
MR_ML	4.22	0.10	1.38	0.91	0.72	0.69	0.28	2.04	1.29	1.77
MR_3A	4.21	0.11	1.12	0.87	0.67	0.56	0.43	1.90	1.23	1.73
FC+FD_2A	4.69	1.28	0.12	1.10	0.54	0.66	0.19	2.17	1.34	2.27
FD+FP_2A	4.80	1.13	0.11	0.98	0.72	0.51	0.37	1.94	1.32	2.29
FC+FP_2A	5.87	1.32	0.24	1.07	0.68	0.96	0.46	1.91	1.56	3.30

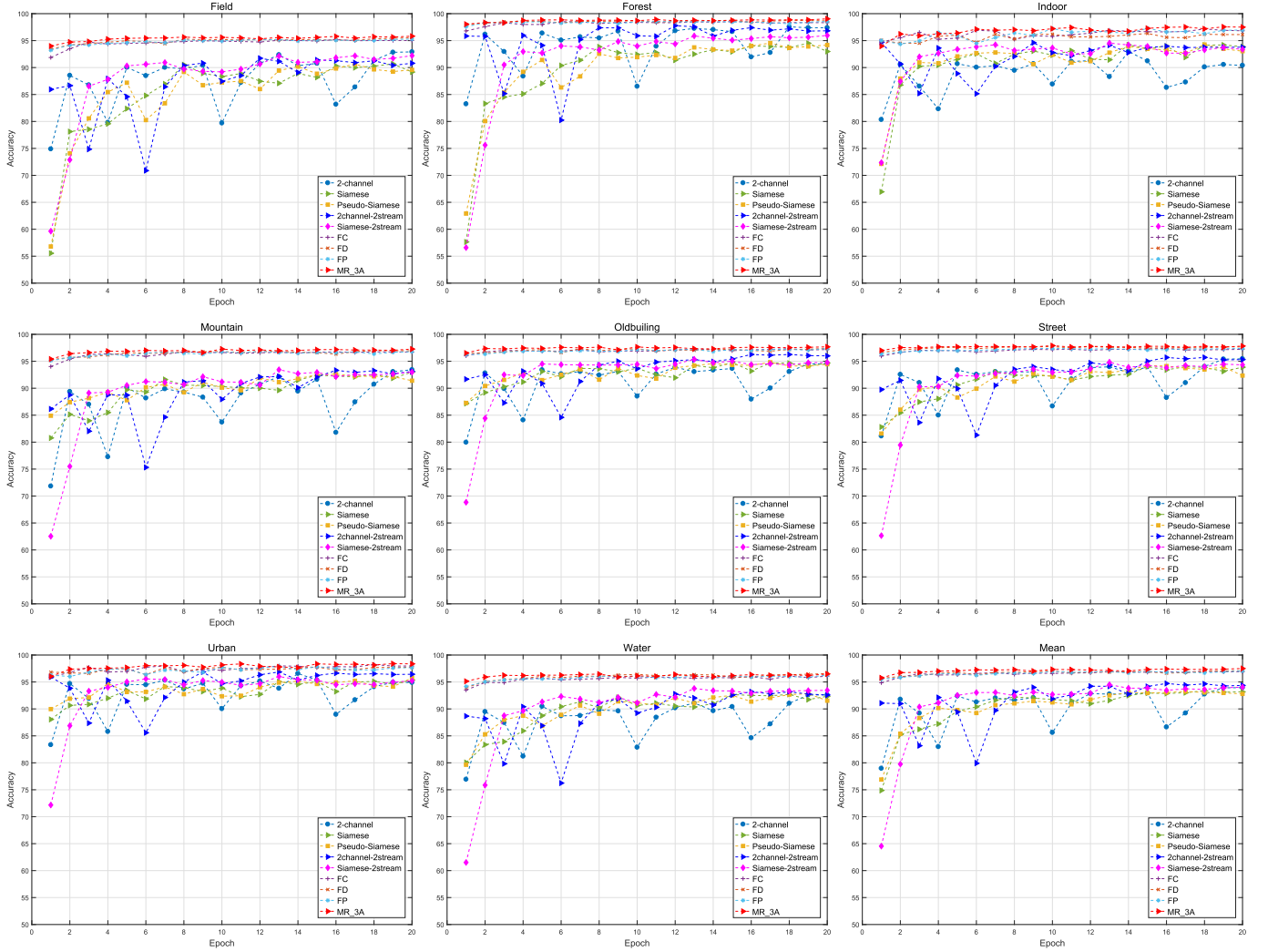


Fig. 11. Matching accuracy of different deep matching methods on multi-modal images, VIS-NIR.

F. Generalization on Multi-Modal Remote Sensing Images

To further verify the effectiveness of our proposal on a more challenging dataset, we tested it on multi-modal remote sensing image patch matching dataset optical-SAR. Figure 13 shows the matching accuracy of various networks

on optical-SAR. Based on the results of Fig. 13, we can reach the following conclusion. Firstly, due to the significant differences between optical and SAR images, the Siamese network with the shared feature extraction branches has the worst matching accuracy. Compared with the Siamese network, the Pseudo-Siamese network with different feature extraction

TABLE VI
MATCHING ACCURACY OF DIFFERENT IMAGE PATCH SIZES ON VIS-NIR DATASET

Size	Field	Forest	Indoor	Mountain	Oldbuilding	Street	Urban	Water	Mean
16×16	90.06	94.83	91.95	91.08	91.90	90.20	94.43	90.60	91.88
32×32	93.46	96.83	94.35	94.60	94.94	94.99	95.96	93.76	94.86
64×64	95.82	99.04	97.50	97.27	97.68	97.83	98.36	96.49	97.50

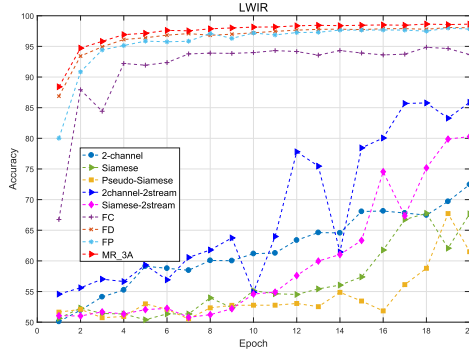


Fig. 12. Matching accuracy of different deep matching methods on multi-modal images, VIS-LWIR.

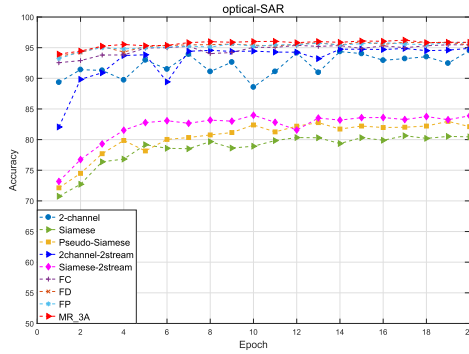


Fig. 13. Matching accuracy of different deep matching methods on multi-modal remote sensing image patch matching dataset, optical-SAR.

branches achieves better results. Secondly, Siamese-2stream network and 2channel-2stream network focus on both the center and surrounding features, which further improve the matching accuracy. Finally, considering the obvious differences between optical and SAR images, we adopt different feature extraction networks in FC, FP, FD and MRAN. Compared with single feature relation learning networks, our proposed MRAN has the best matching result on multi-modal remote sensing image patch matching dataset, which also provides powerful evidence for the effectiveness of our proposal.

G. Feature Relation Activation Analysis

To show the significance of each relation, we compute their mean activation outputs through attention learning (AO) based on 4000 image patch-pairs of VIS-NIR dataset and optical-SAR dataset. From Table. VII we can see that FP has the strongest response for matching patch-pairs and FD has the biggest activation value for non-matching patch-pairs

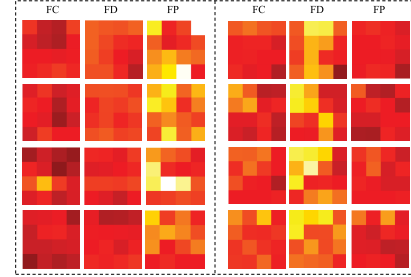


Fig. 14. Feature relation activation map (before attention learning) of four matching patch-pairs (left) and four non-matching patch-pairs (right) on VIS-NIR dataset. The brighter the color, the stronger the response.

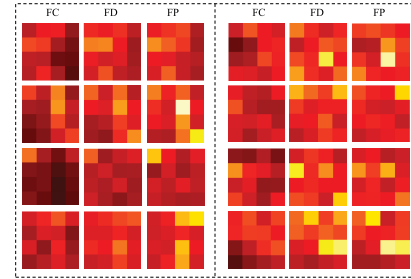


Fig. 15. Feature relation activation map (before attention learning) of four matching patch-pairs (left) and four non-matching patch-pairs (right) on optical-SAR dataset. The brighter the color, the stronger the response.

in each test scenario. Therefore, FP contributes greatly to identifying matching patches and FD has influence on discrimination of non-matching patches on VIS-NIR dataset. Moreover, as shown in Table VIII, FP and FD have similar response while FC gets the smallest activation values on optical-SAR dataset. The phenomena described above also appear in the feature relation activation map (before attention learning), please refer to Fig. 14 and Fig. 15. These results demonstrated that the optimal feature relations are varied for different datasets. So, it is necessary to fuse multiple features relations for more accurate and robust matching.

H. Generalization on Image Patch Retrieval

In this part, we test the generalization of MRAN on the image patch retrieval task. As can be seen from Table IX, MRAN has the highest mean average precision (mAP) on the training dataset, but lower mAP on the test dataset. This observation shows that MRAN falls into over-fitting like other deep methods, AlexNet-conv3 [56], PhilippNet [57] and TFeat [19]. This is because the number of training samples is not enough for training, and there are significant differences

TABLE VII
MEAN ACTIVATION VALUE OF DIFFERENT FEATURE RELATIONS THROUGH ATTENTION LEARNING MODEL (AO) ON VIS-NIR DATASET

Relation	Field	Forest	Indoor	Mountain	Oldbuilding	Street	Urban	Water	Mean
Matching									
FC_AO	0.1578	0.1639	0.1749	0.1605	0.1693	0.1649	0.1707	0.1629	0.1656
FD_AO	0.1677	0.1484	0.1899	0.1751	0.1859	0.1627	0.2011	0.1622	0.1741
FP_AO	0.2139	0.2146	0.2608	0.2233	0.2447	0.2165	0.2682	0.2153	0.2322
Non-matching									
FC_AO	0.1836	0.1898	0.2173	0.1924	0.2086	0.1978	0.2125	0.1925	0.1993
FD_AO	0.2355	0.2385	0.2716	0.2379	0.2609	0.2583	0.2630	0.2467	0.2516
FP_AO	0.1651	0.1644	0.1761	0.1630	0.1672	0.1679	0.1702	0.1645	0.1673

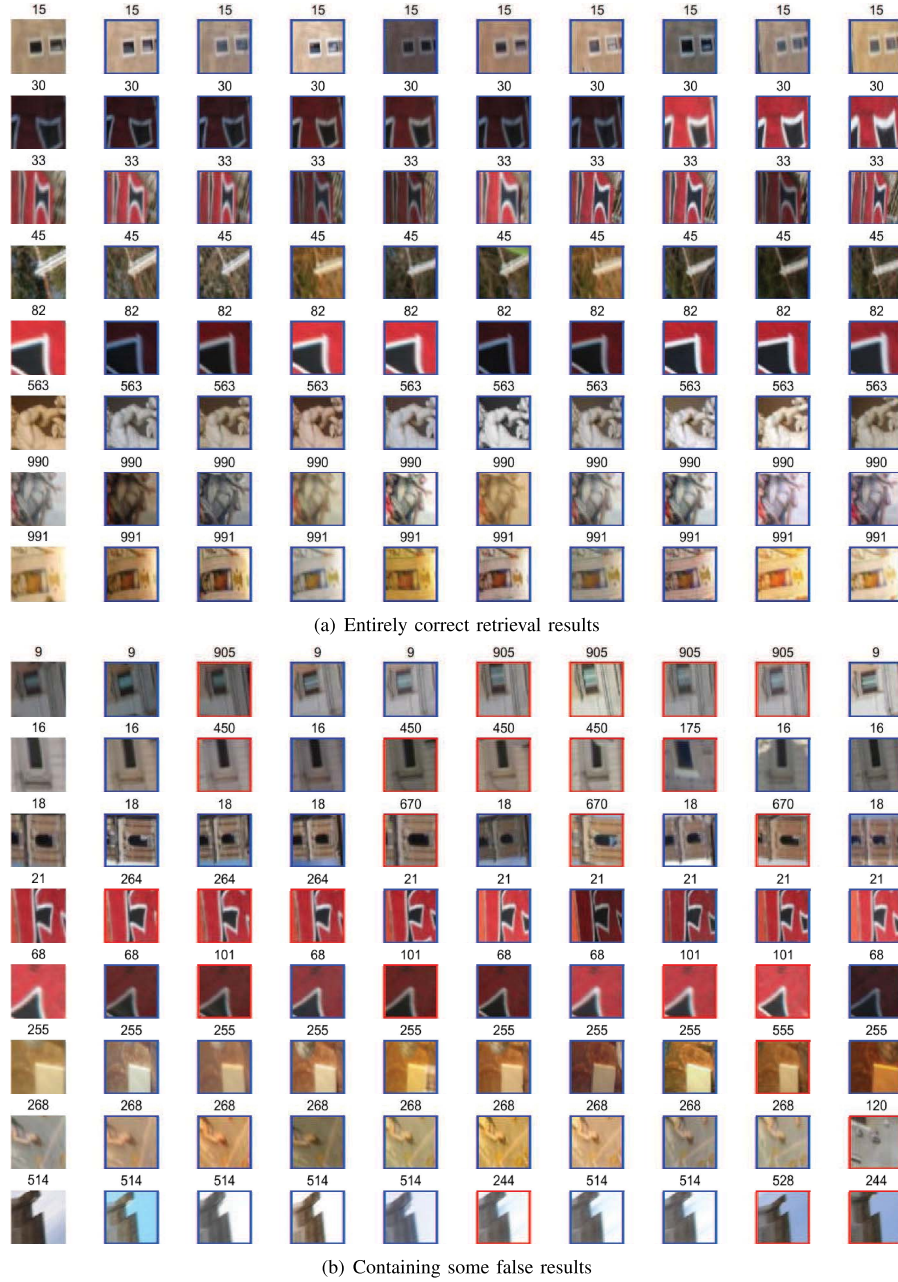


Fig. 16. Image retrieval results on Romepatches dataset. The image patches in the first column represent queries, and the patches in the other columns are the top 9 most similar results retrieved. A patch with a blue border is the correct target, while a patch with a red border is the wrong target. The number above the image patch indicates its 3D point ID.

between the matching patches and high similarity between non-matching patches. Therefore, we transfer the MRAN trained on Yosemite dataset to Romepatches and then fine-tune

it with the Romepatches dataset. The mean result increases from 75.0% to 90.9% after fine-tuning and achieves the best mAP on image patch retrieval.

TABLE VIII
MEAN ACTIVATION VALUE OF DIFFERENT FEATURE RELATIONS
THROUGH ATTENTION LEARNING MODEL (AO) ON
OPTICAL-SAR DATASET

	FC_AO	FD_AO	FP_AO
Matching	0.1357	0.1758	0.1826
Non-matching	0.1456	0.2015	0.1941

TABLE IX
THE MAP(%) RESULTS OF STATE-OF-THE-ART METHODS
FOR IMAGE PATCH RETRIEVAL

Method	Train	Test	Mean
SIFT [11]	91.6	87.9	89.8
AlexNet-conv3 [56]	81.6	79.2	80.4
PhilippNet [57]	86.1	81.4	83.8
TFeat [19]	87.5	81.7	84.6
CKN-raw [58]	79.3	76.3	77.8
CKN-white [58]	91.9	87.7	89.8
CKN-grad [58]	92.5	88.1	90.3
MRAN	93.4	84.4	88.9
MRAN_Yosemite	75.0	75.0	75.0
MRAN_Yosemite_Fine-tune	93.3	88.4	90.9

The retrieval results are shown in Fig. 16. For many image patches, MRAN can find all their targets. However, there are some confusing results as shown in Fig. 16(b), which are very similar to the query image in appearance. Even for humans, it is difficult to distinguish them. Therefore, it is impossible to achieve much more accurate matching results merely relying on image content.

V. CONCLUSION

In this paper, we proposed a multi-relation attention learning network for image patch matching. It takes the complementary advantages of various feature relations to enhance the matching performances and applies an attention mechanism among relations to adaptively learn the fused feature relation. Experimental results demonstrated that our proposal achieves better matching performances, faster convergence speed, and more stable training process. It shows good generalization on multi-modal image patch matching (VIS-NIR, VIS-LWIR), multi-modal remote sensing image patch matching (optical-SAR) and image retrieval tasks. In the future, we will explore how to use the transformer to capture dependencies between local image patches and study more precise image matching with the help of other image patch attributes, such as spatial position, high-order relation and so on.

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